SECTOR RANKING / RESEARCH NOTE

Inside the Sector Champion

Every active factor, exact score weight, and the open-source models and papers behind the research.

Prepared 18 October 2026. This is a concise map of the **locally active** /sector-champion revision dual_w0.01_sip_risk_confirmed0.01_20261017, plus the major models and papers actually reviewed or tested in its research lineage. The score snapshot is 3 October 2026. All 4,436 stocks receive a numerical score; 133 use limited-price-evidence fallback. Historical dates have been repeatedly researched, so the comparisons are not fresh validation.

Exact scoring architecture

Let O be the saved open-model sector percentile, D the saved defensive LightGBM sector percentile, and P the saved price LightGBM sector percentile. The original champion is C = .60 O + .15 D + .25 P. O = .95 L + .05 x mean(CatBoost_defensive, Ridge_defensive), where L is the earlier LightGBM rank_xendcg_balanced_15 challenger. Thus the **expanded coefficients of the original champion** are 57% L, 1.5% CatBoost, 1.5% Ridge, 15% defensive LightGBM, and 25% price LightGBM. These are **component score mixing coefficients**, not learned feature importances or performance attributions. Predictions are normalized to sector percentiles before mixing.

The current website adds:

- H = .01 x (.60 x dual_high_age_percentile + .25 x dual_high_proximity_percentile + .15 x momentum_percentile − .5). Dual means an equal blend of 52-week and available-history recorded closing highs. The maximum score movement is +/-.005. The .60/.25/.15 weights apply **only to H**.
- S = .01 x (sector_SIP_profit_percentile − .5) x invested_fraction x inverse_downside_percentile. The maximum score movement is +/-.005. SIP profit is trailing simulated whole-share gain divided by actual money spent, excluding unused cash. It is not XIRR or expected future return.
- Final score C + H + S; sort descending **within sector**. Unclassified is a display group rather than a verified industry.

Both adjustments abstain if required evidence is missing. High adjustment requires at least 252 observations, 365 history days, 80% active-volume days, recent positive volume and at least five valid peers. SIP also abstains for missing profit, an unresolved relevant corporate action, illiquid/stale evidence or too few peers. They preserve numerical coverage rather than excluding stocks.

Factor inventory: open and defensive models

The earlier LightGBM, CatBoost, Ridge and defensive LightGBM models use these **14 raw variables**: days since recorded high; distance from recorded high; 52-week-high freshness; one-year downside volatility; 12-to-1 momentum; distance from 52-week high; full-history maximum drawdown; months of price history; three-year CAGR; observed five-year CAGR; high support; strict one-year CAGR; six-month momentum; one-year downside frequency. They also use **14 corresponding within-sector percentile variables**, evidence fraction, active-volume fraction and one-year observation count. This is 31 inputs for the open-model family. The defensive LightGBM additionally receives log amount, lookback months, log SIP interval and target CAGR: **35 inputs**. Its saved tree has **zero splits on those four profile fields**. The open CatBoost and Ridge each get only **1.5%** of the original champion's total component mixture.

The earlier LightGBM challenger learned a balanced training label: 50% sector-relative future SIP return rank, 25% drawdown rank, 25% inverse goal-shortfall rank. The CatBoost/Ridge defensive target and defensive LightGBM target used 25% return, 50% drawdown and 25% inverse shortfall. **Training-label weights do not become direct score weights for a feature.**

Factor inventory: price LightGBM

Its **31 raw inputs** are:

- Returns: 21-, 63-, 126- and 252-day return; 12-to-1 momentum.
- High behavior: 63-, 126- and 252-day high gap; 252-day high age; fraction of days near a running high; fraction setting a high; breakout strength.
- Trend: 126-day slope; trend fit (R2); price-path efficiency.
- Risk/distribution: annualized volatility; downside volatility; drawdown; positive-day fraction; unchanged-day fraction; largest-jump share; fifth-percentile daily loss; return skew.
- Moving averages and change: distance from the 21-, 63-, 126- and 252-day moving averages; acceleration.
- Evidence: number of observations; price staleness; history length.

It also receives **31 corresponding within-sector percentile variables**, sector median 126-day return, sector positive-return breadth, stock momentum relative to sector, evidence fraction, and four profile fields. Total: **70 inputs**. The saved model has **zero splits on amount, lookback, interval and target**. The selected rank_xendcg_balanced learning objective concentrated on sector-relative rank, with a target of 50% return, 25% drawdown and 25% inverse shortfall. The website's SIP overlay makes amount, interval and lookback capable of changing final rank; target CAGR affects goal calculations.

Open-source methods in the active score

Method	Active contribution	What we took from it	Boundary
LightGBM and its ranking objectives	57% earlier ranker + 15% defensive ranker + 25% price ranker = 97% of the original champion's mixture	Efficient boosted trees and direct within-sector learning to rank	Trained on our data; no pre-trained stock oracle.
CatBoost, YetiRank	1.5% of original mixture	A distinct ranking algorithm as a small complement	Original CatBoost paper explains the library, not our Indian performance.
Ridge regression	1.5% of original mixture	Regularized linear control alongside trees	Simple statistical model, not a pre-trained financial model.

The percentages above total 100% **before** H and S. The high and SIP adjustments are additive score units, so interpreting them as percentage shares of a nonlinear model would be false.

Papers that directly shaped the active design

Source	Short takeaway	What we actually did
Poh et al., cross-sectional learning to rank	Compare stocks as a ranking problem.	Train/evaluate stock ordering inside sector/date groups. Our SIP targets are project adaptations.
Lyzhin et al., ICML 2023	Ranking loss choice matters.	Compare ranking objectives; retain LightGBM rank_xendcg price and baseline heads.
George and Hwang, 52-week high	Proximity to a rolling high can matter for momentum.	Include high gaps/proximity. The paper does not prescribe our coefficients.
Bhootra and Hur, timing of 52-week high	The age of a rolling high can matter.	Test and retain a small high-recency adjustment by user choice. It does not justify a universal lifetime-ATH penalty.
Moskowitz and Grinblatt, industry momentum	Sector context matters in momentum.	Rank within sectors and add sector median return/breadth context to the price model.
Gu, Kelly and Xiu, asset pricing ML	Nonlinear interactions and characteristics such as momentum/liquidity/volatility warrant testing.	Compare tree and simple linear controls. Their US findings are not claimed as Indian evidence.
Grinblatt and Han, capital gains	Investor gains relative to acquisition cost can affect behavior.	Test a trailing SIP-profit percentile. Our hypothetical ledger is different from the paper's aggregate measure.

Other models and ideas we tested, studied or rejected

These **do not contribute to the current website score** unless their idea is separately noted above. The project reports contain results and limitations; a paper is a hypothesis source, not proof of this model's accuracy.

Method/source	What we learned or tried	Status here
XGBoost learning to rank, KDD paper	Another boosted-tree ranking control, with grouped NDCG training.	Tested as a standalone/blend candidate; not active.
Extra Trees	A randomized-tree control against boosted rankers.	Tested; not active.
TabPFN, official code	Small-data tabular prior and teacher/student ideas.	Direct and hybrid experiments; no broad gate pass; not active.
TabM	Compact neural ensemble specialist.	Tested in specialist rounds; not active.
Explainable Boosting Machine	Additive and limited-interaction specialist, easier to inspect.	Tested; not active.
Microsoft Qlib, DoubleEnsemble	Research pipeline, temporal evaluation, training-sample weighting and feature selection.	Framework/code studied and research experiments run; no Qlib or DoubleEnsemble component in active score.
TRA, DoubleAdapt	Adapt to changing market regimes.	Reviewed; label maturity and historical leakage remain hurdles; not active.
HIST, MASTER, FinGAT	Learn relationships among stocks and sectors.	Reviewed; small peer-context adaptations tested. No graph/attention network in active score.

Method/source	What we learned or tried	Status here
Kronos, Chronos, TimesFM	Pre-trained time-series representation/forecast ideas.	Reviewed as possible challengers; not active. A checkpoint may contain information from dates later than a historical test.
Da, Gurun and Warachka, Frog in the Pan	Smooth progress may differ from one large jump.	Tested continuity proxies such as trend efficiency and jump share; not an extra active overlay.
Daniel and Moskowitz, momentum crashes	Momentum can carry downside risk.	Tested downside/volatility complements; risk variables already exist inside active learned heads.
Safe Policy Improvement, expert aggregation	Be cautious when routing between models with weak evidence or delayed outcomes.	Tested routing/stacking ideas; not active. No formal safety guarantee applies.
Super Learner	Combine out-of-time predictions instead of selecting a single learner.	Tested learned stacking; retained the simpler fixed mixture above.
Open Source Asset Pricing	Catalog published cross-sectional factors with definitions.	Reviewed as a signal reference; its US data/code do not supply Indian historical features.
Quality Minus Junk	Quality screens may help identify risky momentum.	Current-fundamental diagnostic only; undated present-day fundamentals were not inserted into historical training.

The broader literature survey also covers Indian open-source projects (trading_tool, auto-portfolio-management, Nifty-Quant), Alphalens, Zipline, LEAN, portfolio optimizers, FinRL and financial language agents. Those supplied audit, universe or portfolio-workflow ideas. **None is an active ranking component.** They have no numerical weight in this champion.

Reproduction and interpretation

The active package and hashes are in research/runtime/sector-champion-recency/manifest.json. The actual feature lists and training labels are in research/runners/full_coverage_challenger_20261008.py, research/runners/full_coverage_open_models_20261008.py, research/runners/profile_aware_sector_20261008.py and research/runners/sector_price_rebuild_20261010.py. Runtime combination and guard rules are in src/stock_ranker/sector_champion_product.py and src/stock_ranker/sip_profit_signal.py.

The fixed current snapshot is **not a current-market valuation**. Most inputs are price/history-derived; the model does not have a complete point-in-time Indian fundamental, news or valuation record for every stock. None of the cited papers fixes our score coefficients or validates this exact stock universe.